

COMM116

Generative AI Applications

Section 3: World Models and Autonomous Systems

Dr Xiaoyang Wang

Department of Computer Science

x.wang7@exeter.ac.uk

DreamerV3 Revisit

Dreamer V3

Built with the idea of World Models, DreamerV3 has the following features:

- Emphasize “perception” and “representation”;
- Representation enables the agent to imagine the future;
- Incorporates memory.

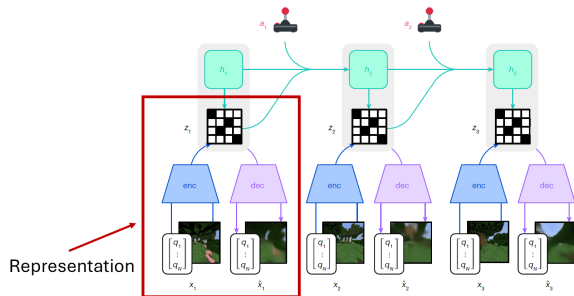
Three neural networks: the **world model**, the **critic** and the **actor**.

Training data: Replayed experience (i.e., trajectories).

Dreamer V3 - World Models

Recurrent state-space model (RSSM)

World model learning



Representation (posterior):

$$z_t \sim q_\phi(\cdot \mid h_t, x_t)$$

- x_t - current frame; “perception”
- h_t - hidden state (memory)

Decoder:

$$\hat{x}_t \sim p_\phi(\cdot \mid h_t, z_t)$$

to ensure informative representations.

x_t could be vectors or image frames.

Hafner, D., Pasukonis, J., Ba, J. and Lillicrap, T., 2025. Mastering diverse control tasks through world models. *Nature*, 640(8059), pp.647-653.



Dreamer V3 - World Models

Recurrent state-space model (RSSM)

The **model state** - $[h_t, z_t]$



Dreamer V3 - World Models

Recurrent state-space model (RSSM)

The **model state** - $[h_t, z_t]$

Use **model state** to

- Predict reward: $\hat{r}_t \sim p_\phi(\cdot \mid h_t, z_t)$
- Predict episode continuation flags: $\hat{c}_t \sim p_\phi(\cdot \mid h_t, z_t) \in \{0, 1\}$
- Reconstruction (see previous slide).

Loss - prediction $\mathcal{L}_{\text{pred}}$:

$$\mathcal{L}_{\text{pred}}(\Phi) := -\log p_\phi(x_t \mid z_t, h_t) - \log p_\phi(r_t \mid z_t, h_t) - \log p_\phi(c_t \mid z_t, h_t)$$



Dreamer V3 - World Models

Recurrent state-space model (RSSM)

The **model state** - $[h_t, z_t]$

Use **model state** to

- Predict reward: $\hat{r}_t \sim p_\phi(\cdot \mid h_t, z_t)$
- Predict episode continuation flags: $\hat{c}_t \sim p_\phi(\cdot \mid h_t, z_t) \in \{0, 1\}$
- Reconstruction (see previous slide).

Loss - prediction $\mathcal{L}_{\text{pred}}$:

$$\mathcal{L}_{\text{pred}}(\Phi) := -\log p_\phi(x_t \mid z_t, h_t) - \log p_\phi(r_t \mid z_t, h_t) - \log p_\phi(c_t \mid z_t, h_t)$$

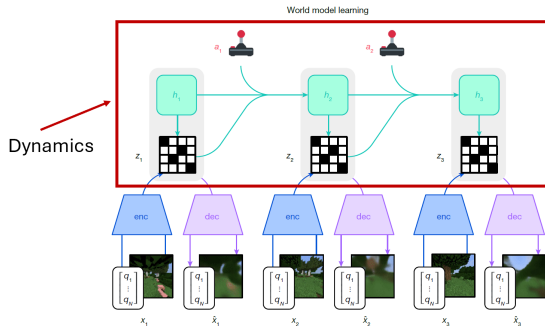
- $-\log p_\phi(x_t \mid z_t, h_t)$ - VAE reconstruction
- $-\log p_\phi(r_t \mid z_t, h_t)$ - reward prediction
- $-\log p_\phi(c_t \mid z_t, h_t)$ - episode continuation prediction

Does not involve **transition dynamics**.

Dreamer V3 - World Models

Recurrent state-space model (RSSM)

Dynamics model.



Transition dynamics in RL:

$$p(s_t | s_{t-1}, a_{t-1})$$

Dynamics model in RSSM world model, operate in latent space z :

$$\hat{z}_t \sim p_\phi(\hat{z}_t | h_t)$$

$$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$$

The difference is involving h_{t-1} in h_t
- **recurrent** memory!

Dreamer V3 - World Models

Recurrent state-space model (RSSM)

Dynamics model $p_\phi(\hat{z}_t | h_t)$ - training

- $\hat{z}_t$ has no ground truth; latent representation.
- Ground truth for perception: x_t .
- The dynamics model's prediction (in latent space) should match the latent representation of ground truth image.

Dreamer V3 - World Models

Recurrent state-space model (RSSM)

Dynamics model $p_\phi(\hat{z}_t | h_t)$ - training

- $\hat{z}_t$ has no ground truth; latent representation.
- Ground truth for perception: x_t .
- The dynamics model's prediction (in latent space) should match the latent representation of ground truth image.

Dynamics loss $\mathcal{L}_{\text{dyn}}$:

$$\mathcal{L}_{\text{dyn}}(\Phi) := \max\left(1, \text{KL}[q_\phi(z_t | h_t, x_t) \parallel p_\phi(z_t | h_t)]\right)$$

Dreamer V3 - World Models

Recurrent state-space model (RSSM)

Dynamics loss $\mathcal{L}_{\text{dyn}}$ - train the dynamics model p_ϕ :

$$\mathcal{L}_{\text{dyn}}(\Phi) := \max\left(1, \text{KL}[\text{sg}(q_\phi(z_t | h_t, x_t)) \parallel p_\phi(z_t | h_t)]\right)$$

Train the representation of the ground truth image (so that representations become more predictable, matches the dynamics prediction):

$$\mathcal{L}_{\text{rep}}(\Phi) := \max\left(1, \text{KL}[q_\phi(z_t | h_t, x_t) \parallel \text{sg}(p_\phi(z_t | h_t))]\right)$$

sg - stop gradient.

$\max(1, KL)$ - if KL is less than 1, clip to be 1; stop minimising KL, thus focusing on other losses.

Dreamer V3 - World Models

Recurrent state-space model (RSSM)

Overall loss:

$$\mathcal{L}_{\text{wm}} = \beta_{\text{pred}} \mathcal{L}_{\text{pred}} + \beta_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \beta_{\text{rep}} \mathcal{L}_{\text{rep}},$$

where it introduces **three hyperparameters**; $\beta_{\text{pred}} = 1$, $\beta_{\text{dyn}} = 1$, $\beta_{\text{rep}} = 0.1$.

- $\mathcal{L}_{\text{rep}}$: Complex scenes contain details unnecessary for control and thus prompt a stronger regularizer to simplify the representations, i.e, larger β ; Simple graphics where individual pixels matter for the task require a weaker regularizer to extract fine details.
- In Dreamer, use a relatively small β_{rep} with the clip function (called “free bits”) solves the problem.
- Fixed hyperparameters across domains (150 diverse tasks).

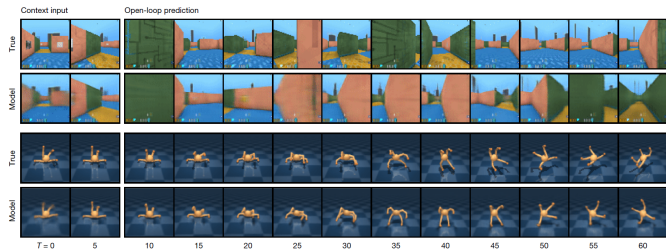
Advantages

Generalisation.

Fixed hyperparameters across domains.

Farsighted planning and learning from **sparse reward**.

Video predictions of the world model



Example: Existing work in autonomous driving

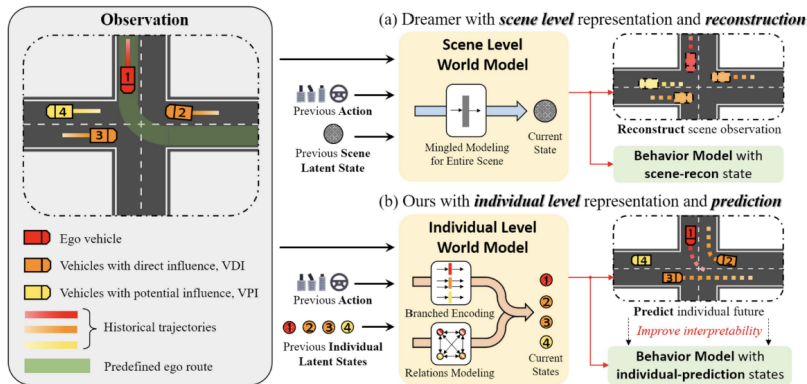
Predictive Individual World Model

Predictive individual world model (PIWM) (Gao et al., 2024)

- In interactive scenerios, such as autonomous vehicles (AVs) in urban scenerios, other road users' intentions are unknown (POMDP).
- World model approaches enable learning within imagination over a compact latent space. The representation is driven by scene-level reconstruction without modeling individual behavior in a scene.
- The transition model in world models do not naturally model interactions.
- Lacks long-term intention modeling; the predictive transition model might not be enough.

Gao, Y., Zhang, Q., Ding, D.W. and Zhao, D., 2024. Dream to Drive With Predictive Individual World Model. IEEE Transactions on Intelligent Vehicles.

Predictive Individual World Model



Gao, Y., Zhang, Q., Ding, D.W. and Zhao, D., 2024. Dream to Drive With Predictive Individual World Model. IEEE Transactions on Intelligent Vehicles.

Predictive Individual World Model

Predictive individual world model (PIWM) (Gao et al., 2024)

Q1: For interactive scenarios, with the transition model, how to encode interactions?

Individual-level transition modeling and **self-attention**.

Recall in Dreamer,

- Transition model (prior): $\hat{z}_t \sim p_\phi(\cdot \mid h_t)$.

With individual world models, the transition is

- $\hat{z}_t \sim p_\phi(\cdot \mid h_t, \text{self_attention})$.

Here the self_attention is calculated from the recurrent states (h_t) of all individuals.

Equivalent to maintaining individual “worlds” for interactive agents.

Gao, Y., Zhang, Q., Ding, D.W. and Zhao, D., 2024. Dream to Drive With Predictive Individual World Model. IEEE Transactions on Intelligent Vehicles.

Predictive Individual World Model

Predictive individual world model (PIWM) (Gao et al., 2024)

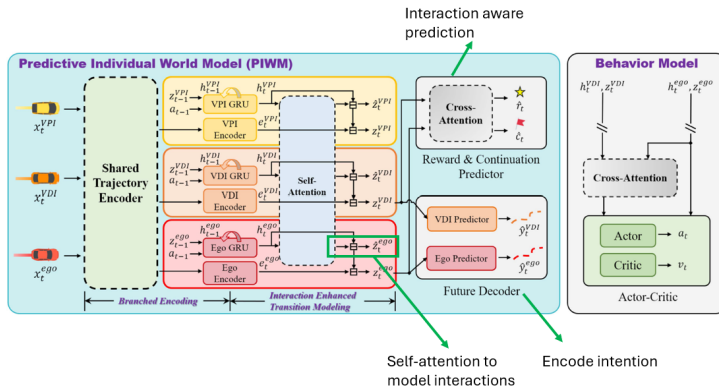
Q2: The latent state in world models contains memory of previous states and actions (*the history*). Could it also model intention (*the future*)?

Future decoding: use decoder to predict future trajectories over a 2-second horizon, forcing the latent state to implicitly learn about the intention.

Intention of surrounding agents are implicitly fed into the latent state via self-attention.

Predictive Individual World Model

Predictive individual world model (PIWM) (Gao et al., 2024)



Interaction-aware individual world model. Figure adapted from Gao et al., 2024.

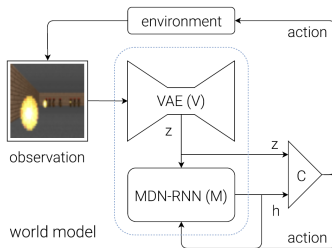
Challenges

- Model intentions and preferences of interactive agents. Latent representation for human behaviour?
- Beyond autonomous driving - collaborative human-robot interaction, and other tasks?



Figure from Habitat 3, <https://aihabitat.org/habitat3/>.

Challenges



Ha, D. & Schmidhuber, J. (2018).

- Is it the best way to design an agent with a mental model and memory, with the ability to learn and interact with the external world?

Summary



Summary

- MBRL.
- World models, learning with imagination.
- Applications: autonomous driving, autonomous robots and beyond.

References

- Ha, D. and Schmidhuber, J., 2018. Recurrent world models facilitate policy evolution. Advances in neural information processing systems, 31.
- Luo, F.M., Xu, T., Lai, H., Chen, X.H., Zhang, W. and Yu, Y., 2024. A survey on model-based reinforcement learning. Science China Information Sciences, 67(2), p.121101.
- Hafner, D., Lillicrap, T., Fischer, I., Villegas, R., Ha, D., Lee, H. and Davidson, J., 2019, May. Learning latent dynamics for planning from pixels. In International conference on machine learning (pp. 2555-2565). PMLR.
- Hafner, D., Pasukonis, J., Ba, J. and Lillicrap, T., 2023. Mastering diverse domains through world models. arXiv preprint arXiv:2301.04104.
- Sundar, A., Luo, C. and Wang, X., 2025. Enter the Void: Exploring with High Entropy Plans. arXiv preprint arXiv:2505.16787. (Under Submission)
- Gao, Y., Zhang, Q., Ding, D.W. and Zhao, D., 2024. Dream to Drive With Predictive Individual World Model. IEEE Transactions on Intelligent Vehicles.

Discussions

Paper 1: Bar, A., Zhou, G., Tran, D., Darrell, T. and LeCun, Y., 2025. Navigation world models. In Proceedings of the Computer Vision and Pattern Recognition Conference (pp. 15791-15801).

Paper 2: Gao, S., Zhou, S., Du, Y., Zhang, J. and Gan, C., 2025. Adaworld: Learning adaptable world models with latent actions. arXiv preprint arXiv:2503.18938.